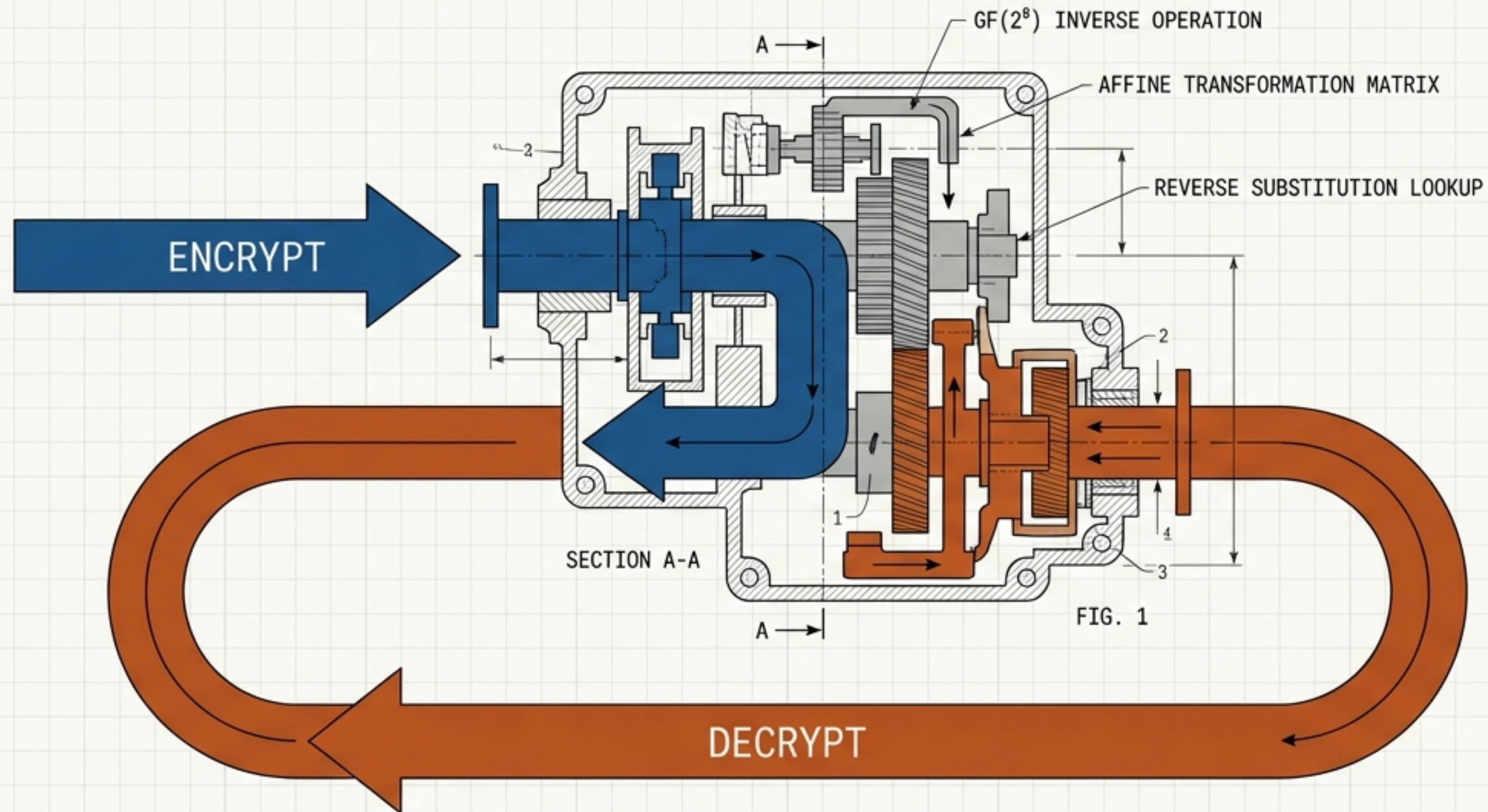


THE MECHANICAL BLUEPRINT OF THE AES INVERSE S-BOX

Deconstructing the mathematics of cryptographic reversibility.



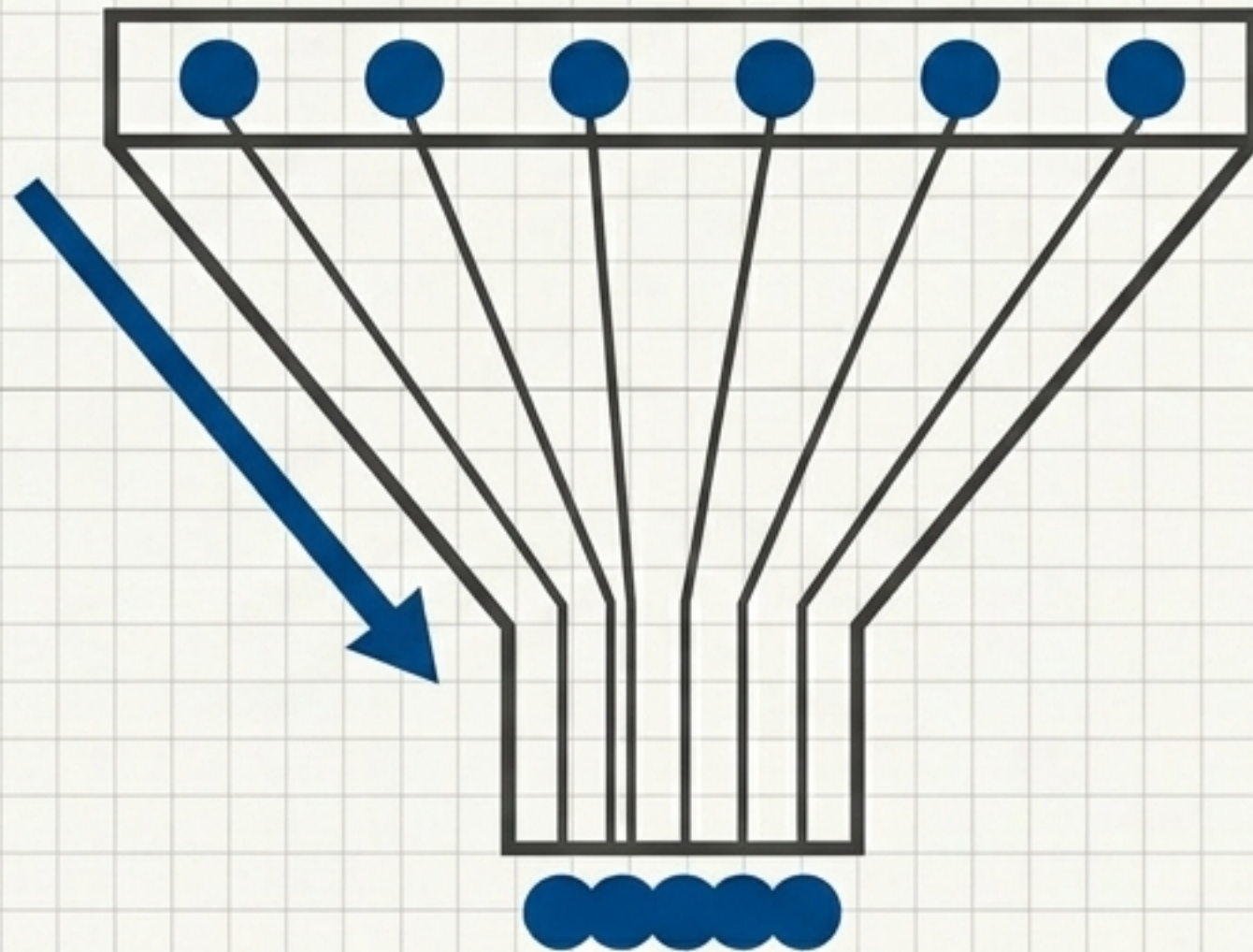
Encryption without decryption is simply data destruction.



The S-box provides nonlinearity (confusion) to the AES block cipher. However, it cannot merely scramble bytes in a one-way fashion. It must preserve structure underneath the confusion so authorized users can walk backward through the machine.

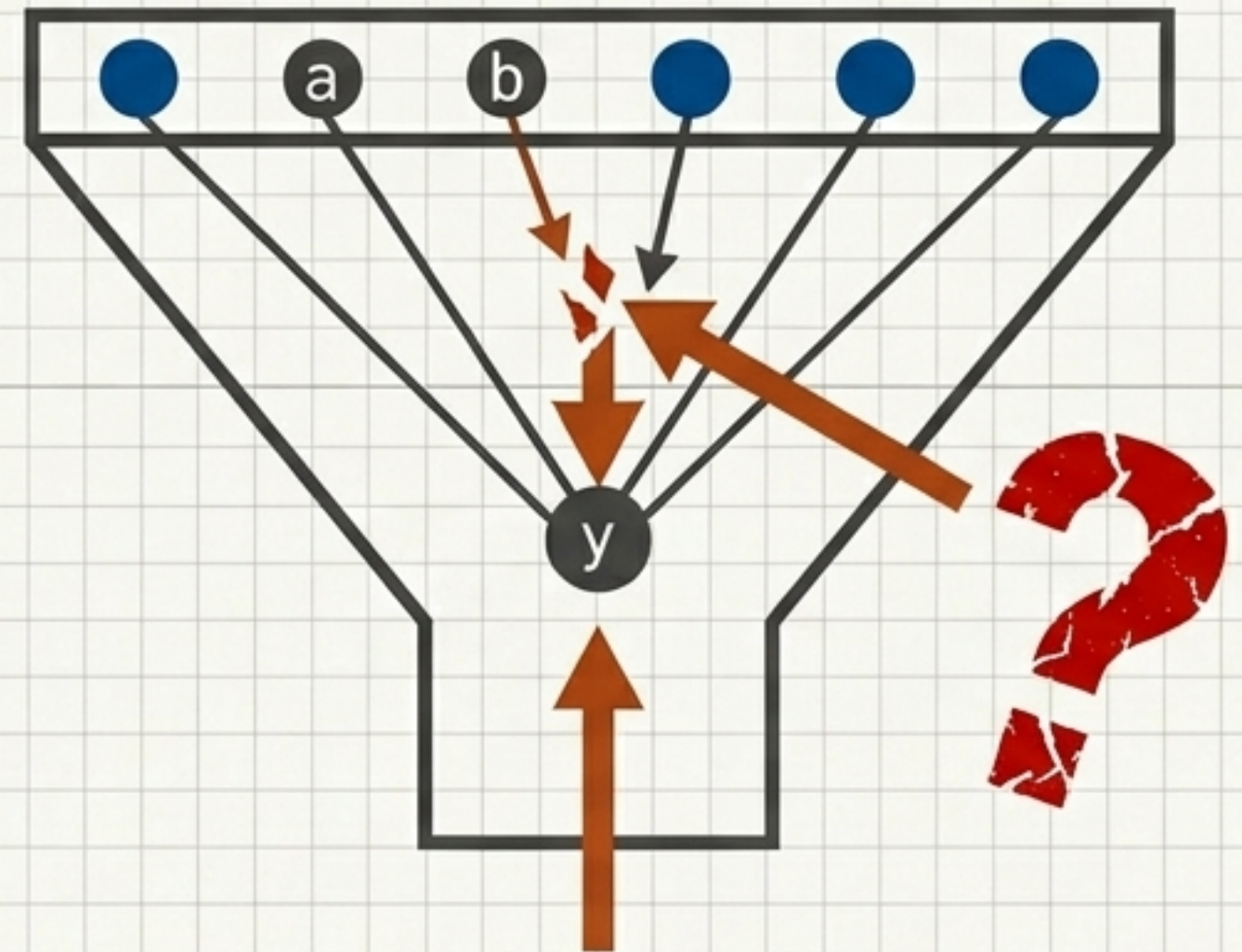
Valid vs. Invalid Encryption Mappings

Valid: Bijective Mapping (Permutation)



Every input has exactly one output, and every output comes from exactly one input.

Invalid: Ambiguity



If $S(a) = y$ and $S(b) = y$, the inverse machine cannot know which path to recover. Information is permanently lost.

Forward direction



To get ready, operations happen in a strict sequence.

Reverse direction



To reverse a sequence of operations, undo them in the opposite order. Untie first, remove shoes second.

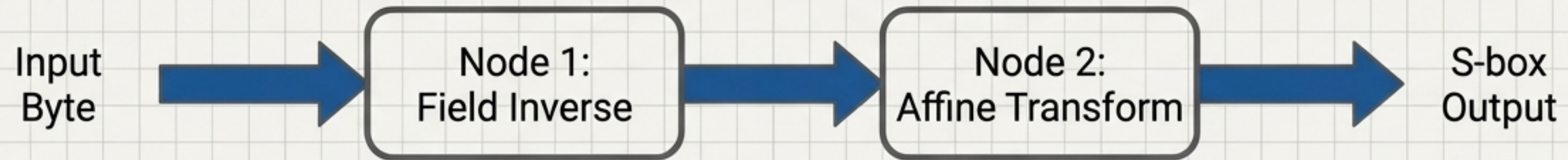
Reversibility is a matter of walking backward through the machine.

S-box Structure: Forward vs. Reverse Pipeline

Reversibility is achieved by undoing operations in the opposite order through the machine.

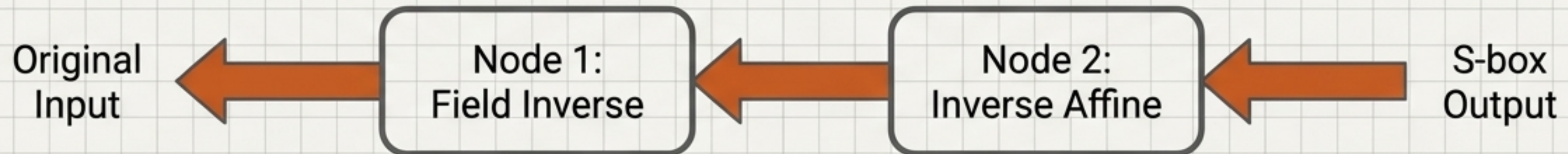
Forward direction (Encryption)

$$S(a) = A(a^{-1})$$



Reverse direction (Decryption)

$$S^{-1}(y) = (A^{-1}(y))^{-1}$$



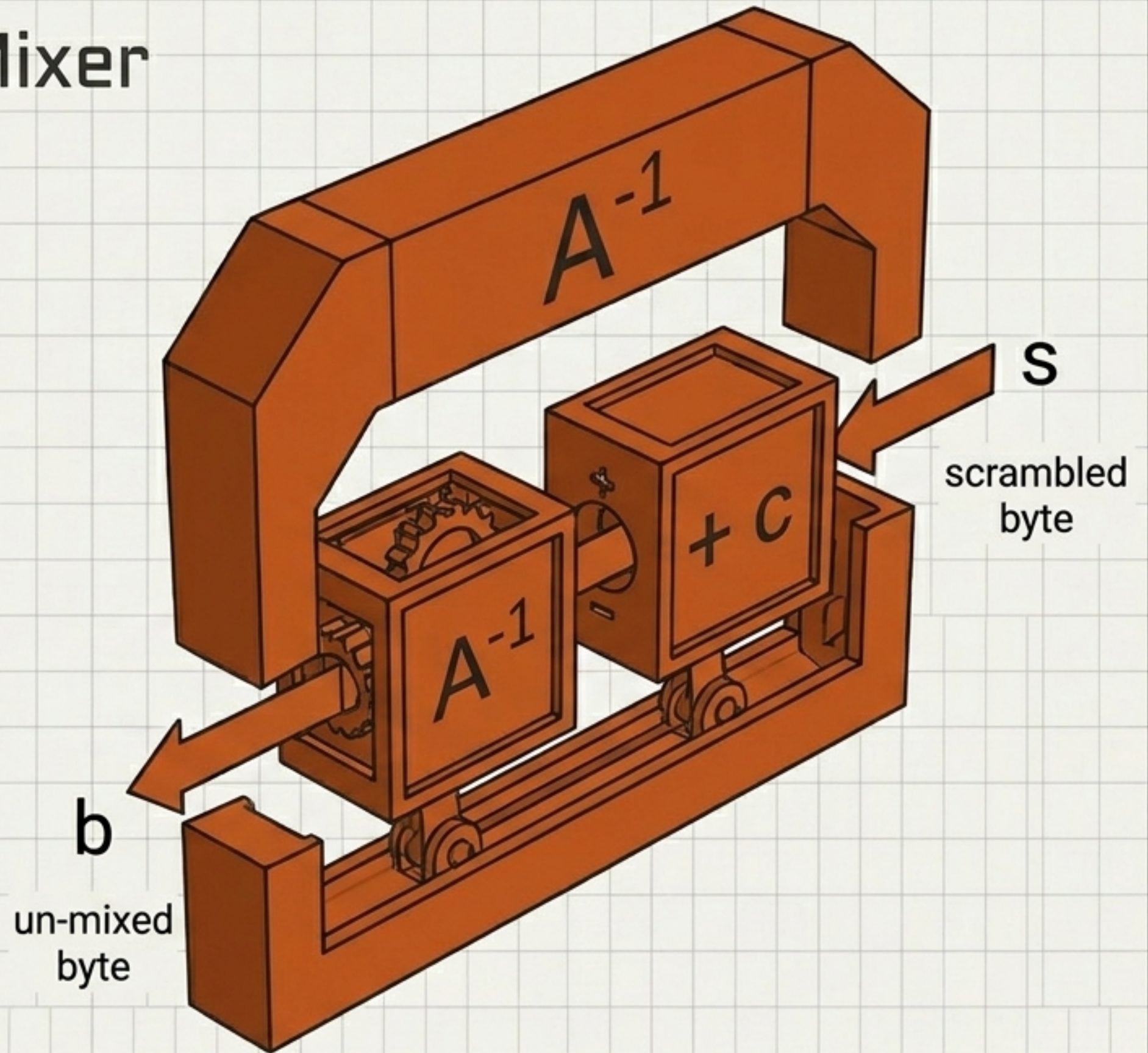
The Mathematical Un-Mixer

- The forward affine step is a bit mixer with a constant:
$$s = Ab + c$$
- The inverse affine step is mathematically paired to unwind it:

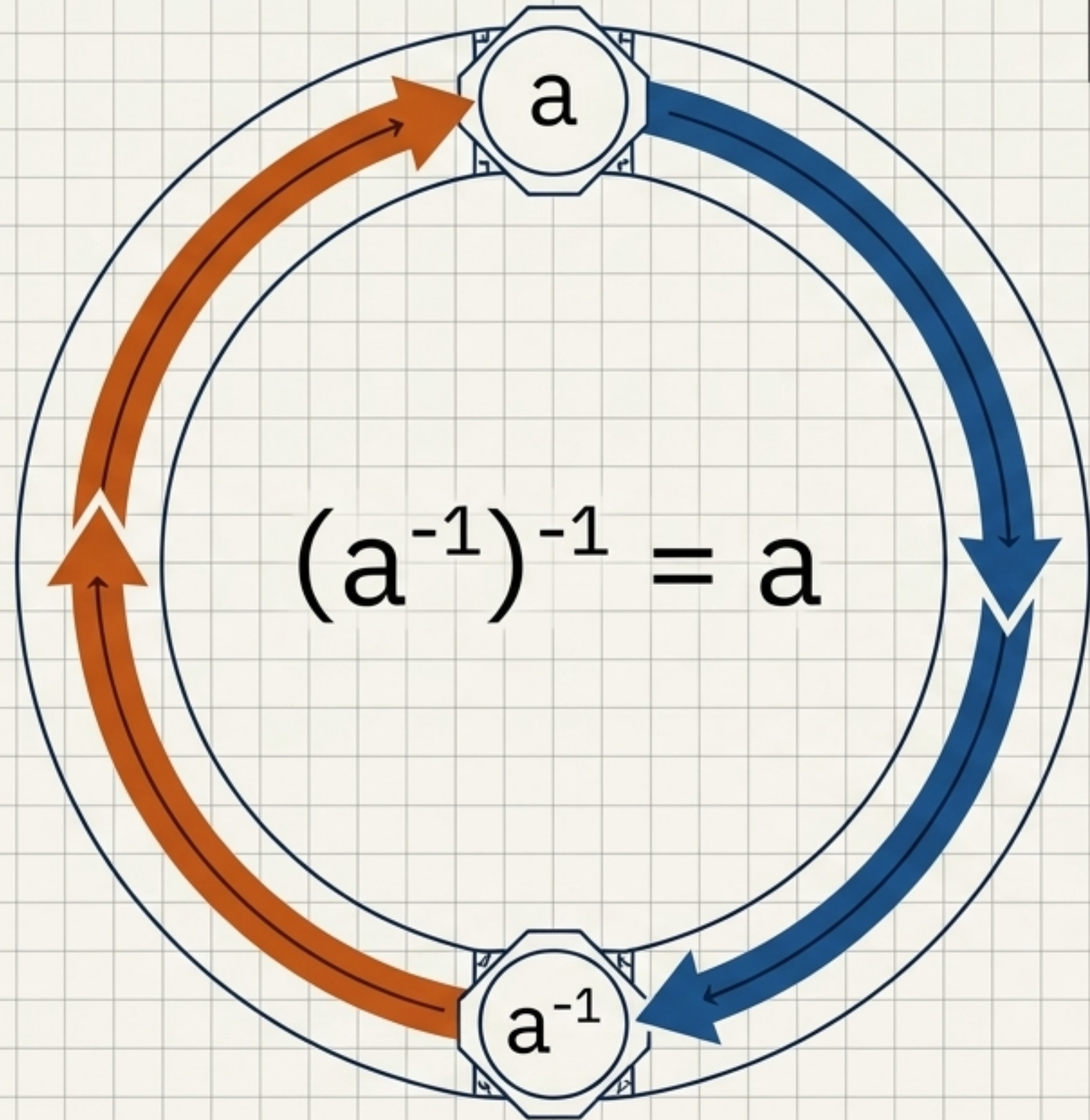
$$b = A^{-1}(s + c)$$

Why Addition?

In the finite field $GF(2)$, subtraction and addition are identical (XOR). Adding the constant 'c' again is the exact mathematical equivalent of subtracting it.



The Self-Undoing Operation



The Self-Undoing Operation

In a finite field, taking the multiplicative inverse is its own undoing.

We do not need a new mathematical operation to reverse this step; applying the exact same operation twice returns the original value.

The Zero Edge Case

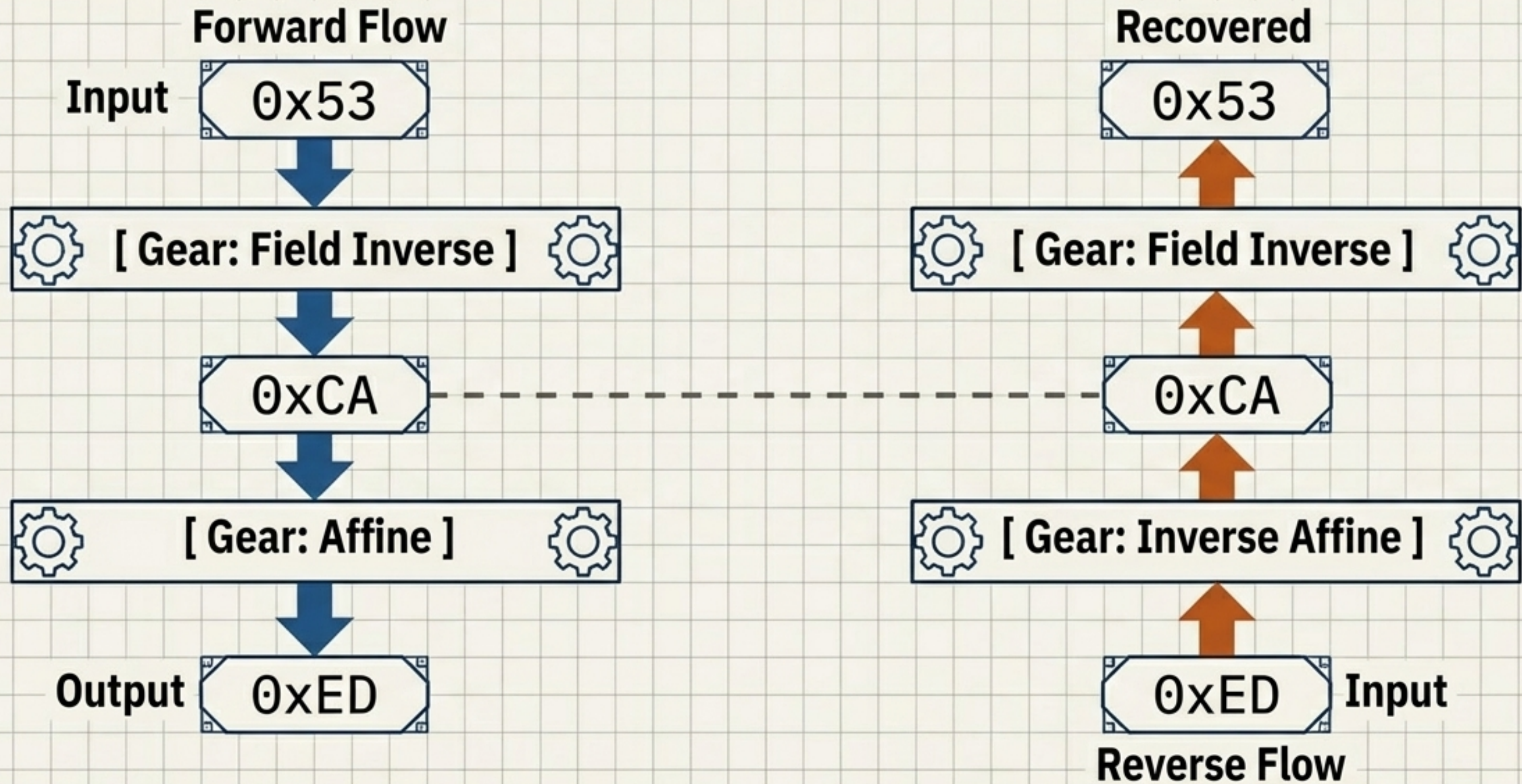
AES handles 0x00 specially. Since zero has no multiplicative inverse, it maps to itself. Zero goes to zero in the forward step, and maps back consistently in the inverse step.

The Forward vs. Reverse Architecture Matrix

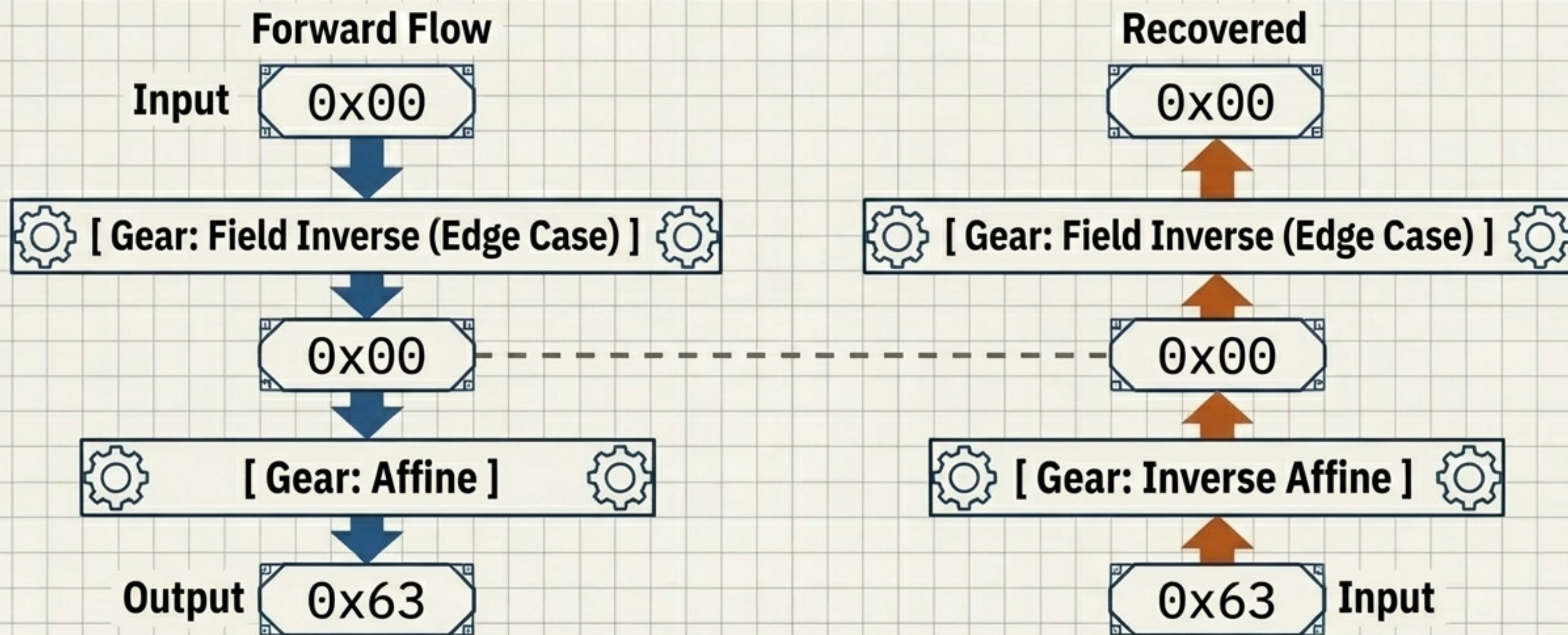
Phase	Direction	Stage 1 Operation	Stage 2 Operation	Core Formula
Forward S-Box	Left-to-Right	Field Inverse	Affine Transform	$S(a) = A(a^{-1})$
Inverse S-Box	Right-to-Left	Inverse Affine	Field Inverse	$S^{-1}(y) = (A^{-1}(y))^{-1}$

Note that while the field inverse is its own undoing, the full S-Box requires a separate **Inverse Affine** step. The entire S-box is NOT its own inverse.

Forward and Reverse Transformation Synchronization



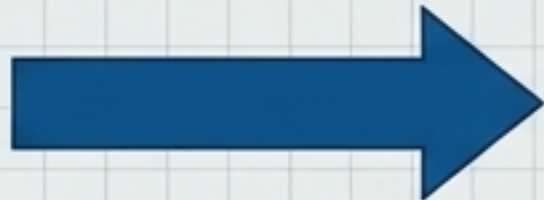
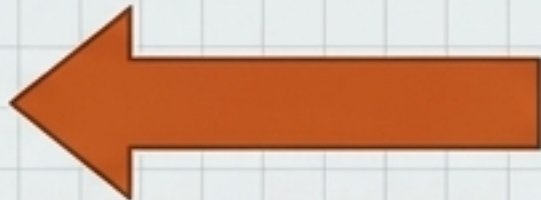
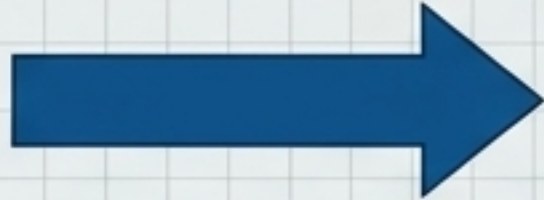
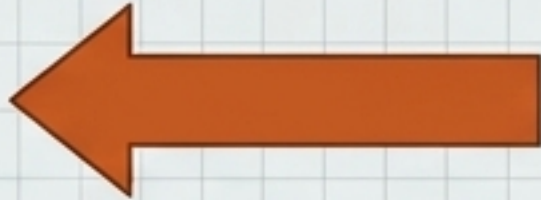
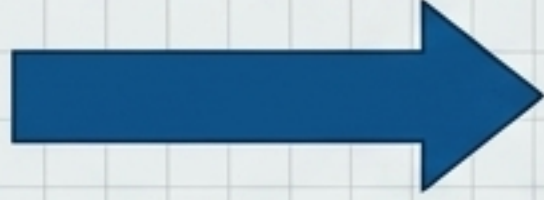
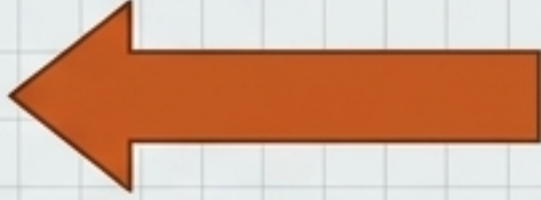
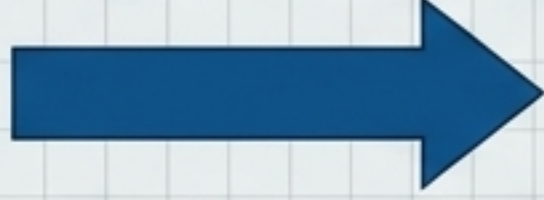
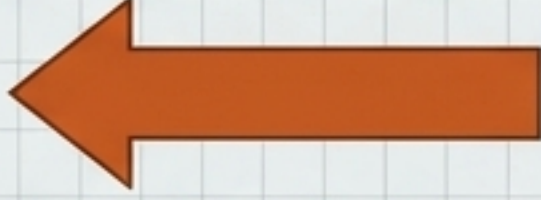
Forward and Reverse Transformation Synchronization (Edge Case 0x00)



Even the artificial exception for zero survives the reverse pipeline perfectly: $S^{-1}(0x63) = 0x00$.

The Dual-Read Table (2-bit Toy Example)

Read Left-to-Right (Encrypt).
Find input on left, output
output is on the right.

00			10
01			11
10			01
11			00

Read Right-to-Left (Decrypt).
Find input on right, output,
output is on the left.

The inverse table is not a second unrelated mathematical entity.
It is simply the forward table read backward. Swapping
input-output relationships is all it takes to build it.

$$S[0x00] = 0x63 \iff \text{InvS}[0x63] = 0x00$$

$$S[0x53] = 0xED \iff \text{InvS}[0xED] = 0x53$$

$$S[0x7C] = 0x10 \iff \text{InvS}[0x10] = 0x7C$$

$$S(x) = y \iff S^{-1}(y) = x$$

If the forward table assigns an output, the inverse table **MUST** flawlessly reflect that mapping to ensure zero data loss.

Troubleshooting Cognitive Friction: Myth vs. Mathematics

Myth	Mathematics
The full S-box is its own inverse.	Only the field inverse step is its own undoing. The affine transformation changes the composition, requiring a dedicated inverse procedure.
You reverse the steps in the same order they were applied.	Compositional reversal requires opposite order: Inverse Affine MUST happen before the Field Inverse again.
The Inverse S-box is an arbitrary scrambling algorithm.	It is strictly determined by the forward S-box. It is the exact mathematical negative space.

Master Formula: $S^{-1}(y) = (A^{-1}(y))^{-1}$

InvS[0xED] = 0x53

0x53

[Field Inverse]

[Inverse Affine]

0xED

Confusion Without Destruction

The S-box gives AES nonlinearity to resist cryptanalysis, but through bijections and perfectly paired inverse operations, it preserves absolute mathematical integrity so decryption can flawlessly recover the original data.